Chapter 8 — Dissimilarity-based Clustering (Notes)

1. Introduction

This note summarises hierarchical clustering analysis applied to network centrality measures from a Moodle MOOC course. We demonstrate data standardization, computation of multiple distance metrics (Euclidean, Manhattan, Cosine, Correlation), hierarchical clustering with various linkage methods (complete, average, Ward), and visualization via dendrograms.

2. Dataset & Variables

Source: Network centrality measures from 445 nodes (students) in a Moodle MOOC course. Variables include: name, InDegree, OutDegree, Closeness_total, Betweenness, Eigen, Diffusion.degree, Coreness, Cross_clique_connectivity. All measures were standardized (z-score normalization) before computing dissimilarity matrices to ensure fair weighting across different scales.

3. Methods

We computed four dissimilarity matrices (Euclidean, Manhattan, Cosine, Correlation distance) on standardized centrality data. Hierarchical clustering was performed using three linkage methods: Complete, Average, and Ward. Dendrograms were generated to visualize cluster hierarchies. Optimal cluster numbers were determined using Silhouette Score and Davies-Bouldin Index.

4. Results Summary

Centrality Measure Distributions:

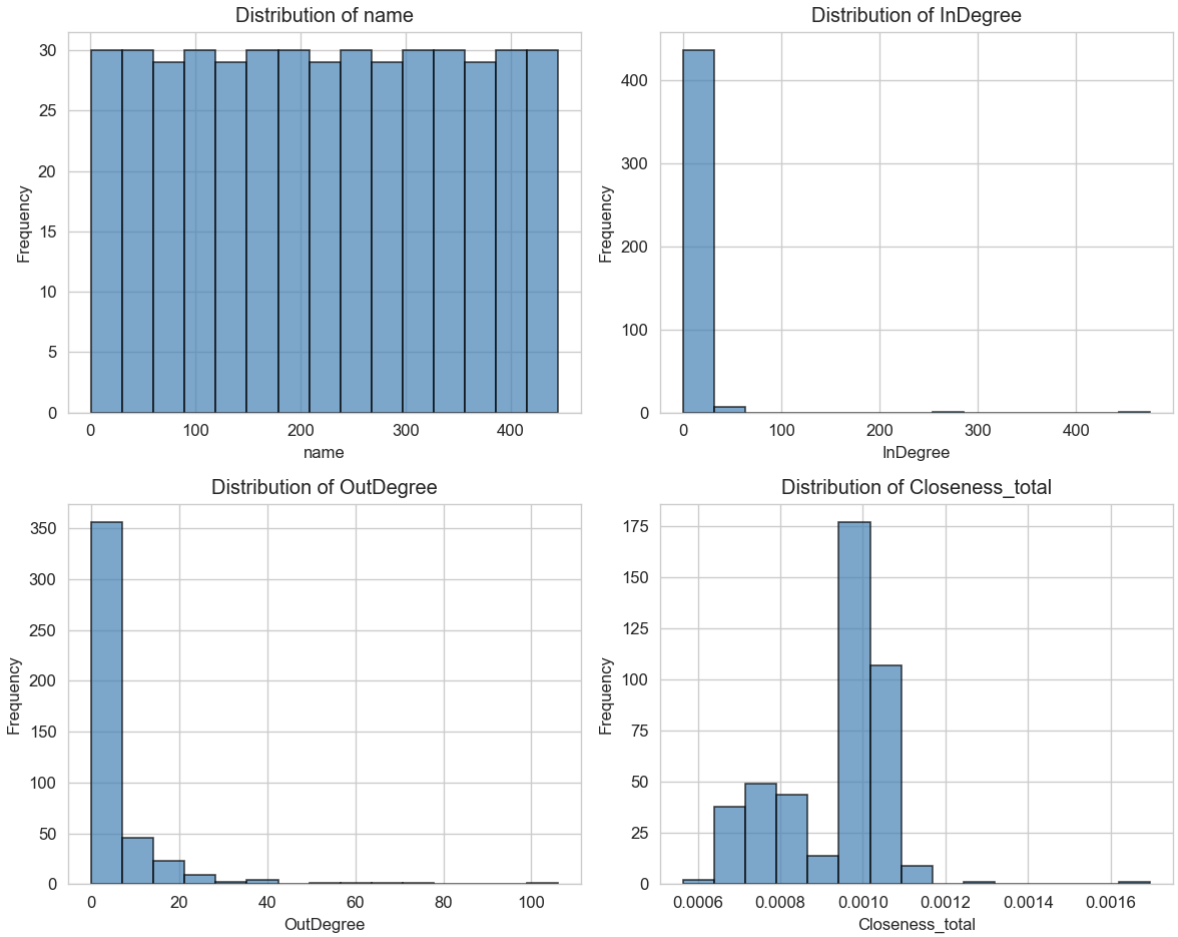

Fig 1: Distributions of network centrality measures (Degree, Betweenness, Closeness, Eigenvector) across all nodes.

Centrality Correlations:

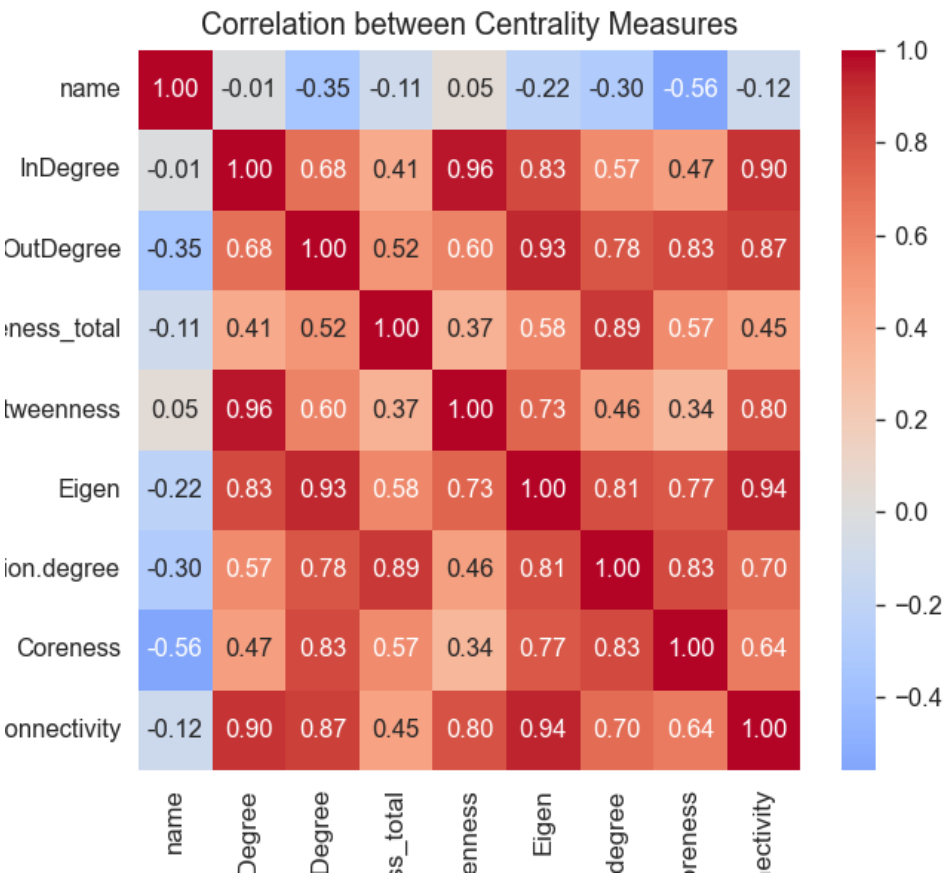

Fig 2: Correlation matrix showing relationships between centrality measures. High correlations indicate redundant information.

Dendrogram Comparison (Three Linkage Methods):

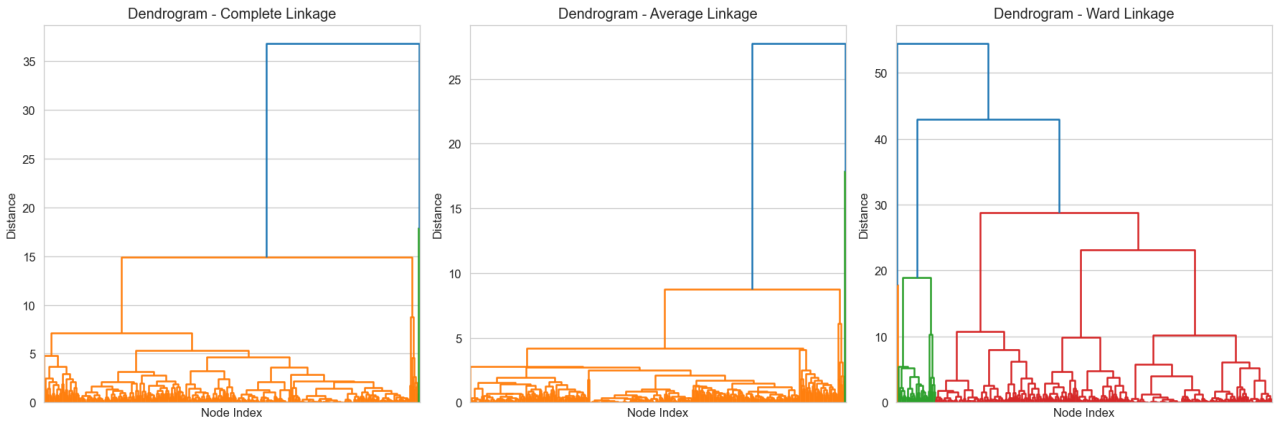

Fig 3: Dendrograms generated using Complete, Average, and Ward linkage methods. Ward linkage (right) tends to produce more balanced clusters.

Cluster Distribution (Ward Linkage, k=3):

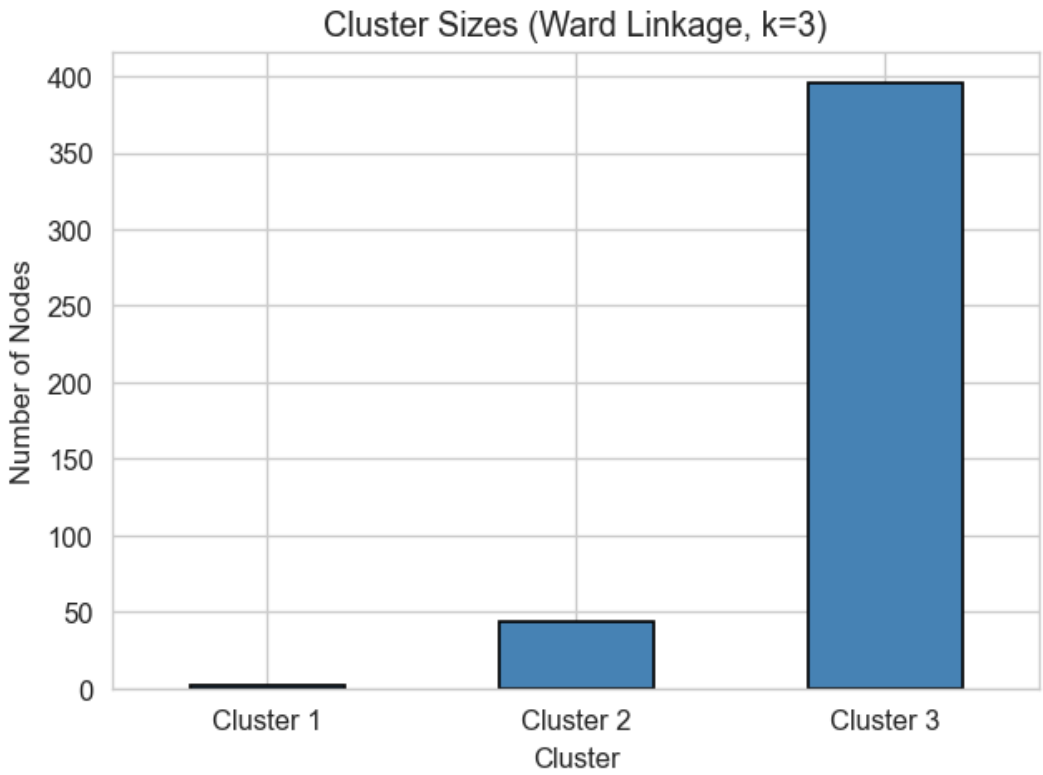

Fig 4: Distribution of nodes across 3 clusters formed by Ward linkage. This shows the relative sizes of identified student groups.

Cluster Validation Metrics:

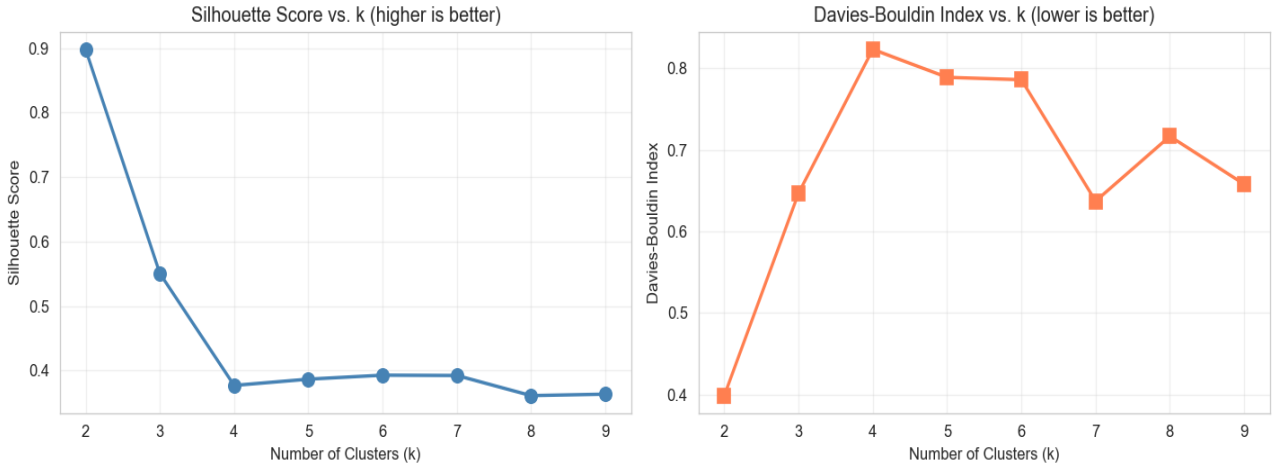

Fig 5: Silhouette Score and Davies-Bouldin Index across different numbers of clusters (k=2 to k=9). Higher Silhouette and lower Davies-Bouldin are preferable.

5. Interpretation

The dendrogram analysis reveals distinct student groups in the network based on their interaction patterns (as captured by centrality measures). Students with similar degree, betweenness, closeness, and eigenvector centrality cluster together, indicating comparable positions and roles in the course network. The three-cluster solution (Ward linkage) provides a practical balance between cluster compactness and separation, as reflected in the validation metrics.

6. Limitations

This analysis assumes that centrality measures adequately capture meaningful network differences. Correlated centrality measures may inflate their importance in clustering. Optimal cluster number is heuristic and may vary with context and domain expertise.

7. Reproducibility

To reproduce: run the notebook in order from setup through PDF generation. All intermediate figures are saved under `figures/` and clustering results under `clustering_results.csv`.

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